Observational Evidence of Intrinsic Differential Redshifts in Galaxy Pairs: A Test of the Decreasing Universe Theory (DUT)

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Abstract

Conventional cosmological models based on the FLRW metric predict that the distribution of relative redshifts between satellite galaxies and their primary components in virialized groups must be perfectly symmetric, reflecting only isotropic peculiar velocities (Doppler effect). In this work, we analyze a highly controlled sample of $n = 700$ *galaxy pairs in dense environments (including the Virgo, Fornax, and Hydra clusters). The data reveal a persistent systematic asymmetry, where the massive primary component exhibits a lower redshift than predicted in* 56.28% *of the cases. The statistical hypothesis test rejects random symmetry with a Z-score of* 3.33 *and a p-value of* 0.00043 ***(*$\boldsymbol{99.957\%}$ *confidence level**). We demonstrate that this result elegantly corroborates the predictions of the Decreasing Universe Theory (DUT), in which the differential metric shrinkage of matter—modulated by local mass density and structural evolutionary time—generates intrinsic non-Doppler redshifts.*

1. Introduction

In the standard cosmological model (ΛCDM), the cosmological redshift (z) is interpreted as the expansion of the very fabric of space-time. When analyzing geometrically connected and gravitationally bound galaxy systems (such as binary pairs and dwarf satellites), the cosmological redshift is expected to be identical for both bodies, and the only observed fluctuations should arise from orbital motion around the common center of mass. This premise demands a rigorously symmetric distribution of relative redshifts and blueshifts along a statistical line of sight.

However, systematic discrepancies in satellite kinematics have been frequently reported in the astrophysical literature, often treated as observational anomalies. This paper proposes a radically different approach: the interpretation of these shifts in light of the Decreasing Universe Theory (DUT), proposed by João Holland Barcellos. The DUT postulates that the universe is not spatially expanding; instead, space and matter are undergoing a homogeneous and continuous metric contraction over cosmological time.

2. The DUT Scenario and the Differential Shrinkage Hypothesis

In the DUT mathematical formulation, the observed redshift (z) reflects the shrinkage of the observer's atoms between the moment of emission and reception. Extending this to dense local systems, we propose two postulates:

Cumulative Temporal Effect: Cosmically older structures undergo the metric contraction process for a longer duration, accumulating a smaller scale factor.

Mass Density Modulation: More intense gravitational fields accelerate the local rate of matter shrinkage.

If a primary galaxy (more massive and older) shrinks faster than its satellite, its atoms today will be intrinsically smaller. Consequently, light emitted by the primary galaxy will experience a relative blueshift compared to the satellite, resulting in a systematically lower redshift for the dominant component of the pair.

3. Methodology and Observational Data

We consolidated a database of 700 galaxy pairs ($n = 700$), divided into 35 sampling blocks. Criteria included:

Strict spatial proximity and proven gravitational binding.

Distance data calibrated by redshift-independent methods (TRGB, Cepheids, and SBF).

High-precision spectroscopic measurements for the primary (z_1) and secondary (z_2) components.

The binary variable Criterion was defined as 1 where the massive primary displayed a lower redshift than its satellite ($z_1 < z_2$), and 0 for the inverse.

4. Statistical Analysis and Results

Unifying the 35 sample blocks yielded:

Total number of pairs (n): 700

Criterion successes ($z_1 < z_2$): 394 pairs

Observed proportion ($\hat{p}$): 56.2857%

Applying the hypothesis test for proportions ($H_0: p = 0.50$):

Z-score: 3.33
p-value: 0.00043 (**99.957% Confidence Level**).

4.1-Analysis in relation to the average of the measurements:

An additional analysis was performed to investigate whether pairs classified as successes (Criterion = 1) exhibited systematically larger differences in stellar age and mass than pairs classified as failures (Criterion = 0).

For each pair, the difference in mean stellar age ($\Delta Age = Age_1 - Age_2$) and the mass ratio ((M_1/M_2)) were calculated.

The results show that successful pairs display, on average, greater disparities between the galaxies than failed pairs.

The mean age difference was 5.47 Gyr for the successful pairs, compared to 5.21 Gyr for the failed pairs.

Similarly, the median mass ratio was 131 for the successful pairs and 92 for the failed pairs.

Although these differences are moderate, both trends are in the direction predicted by the Decreasing Universe Theory (DUT), according to which older and more massive galaxies should tend to exhibit a stronger effect on the observed redshift.

In particular, *the mass contrast appears more pronounced than the age contrast*, suggesting that mass may play a more significant role in the observed phenomenon.

These results reveal a statistical association that is consistent with the qualitative predictions of the DUT.

4.2- Combined Influence of Mass and Age

In order to investigate whether the combination of mass and age provides greater discriminatory power than either parameter alone, the following quantity was defined:

P = M_1 * Age_1 - M_2 * Age_2

where (M_1) and (Age_1) correspond to the more massive and older galaxy in the pair, while (M_2) and (Age_2) correspond to the less massive and younger galaxy.

All 700 galaxy pairs were ranked according to the value of (P), and the sample was subdivided based on the highest values of this parameter. The proportion of pairs classified as successful (Criterion = 1) was then calculated for each subset.

Sample	Number of Pairs	Criterion = 1
Full sample	700	56.6%
Top 30% highest values of (P)	210	60.0%
Top 20% highest values of (P)	140	63.6%
Top 10% highest values of (P)	70	64.3%
Top 5% highest values of (P)	35	65.7%

A monotonic increase in the success rate is observed as the combined mass-age contrast between the galaxies increases. While the full sample exhibits a success rate of 56.6%, the subset corresponding to the top 5% highest values of (P) reaches a success rate of 65.7%.

This behavior suggests that the combination of mass and age contains more predictive information than either variable considered separately. Furthermore, the observed trend is qualitatively consistent with the Decreasing Universe Theory (DUT), according to which older and more massive galaxies should tend to exhibit a stronger influence on the observed redshift. This trend does constitute an evidence of a causal relationship, it indicates that the combined contrast in mass and age is statistically associated with a higher frequency of the outcome predicted by the DUT.

An additional noteworthy feature is that the increase in the success rate is gradual and monotonic across all subsets analyzed (56.6% → 60.0% → 63.6% → 64.3% → 65.7%), which strengthens the evidence that the observed effect is not driven solely by a small number of extreme cases.

5. Discussion: Confronting ΛCDM vs. DUT

The DUT provides a direct and intrinsic explanation for this asymmetry. The primary galaxy, possessing orders of magnitude more mass and a more evolved core, contracts at a slightly higher rate than its dwarf companions. This gradient in atomic scale shifts the primary's spectrum toward a relative blueshift.

However, a fundamental question arises: *why does this asymmetry stabilize at* $\approx$ 56% *rather than a much higher index?* The answer lies in the *Peculiar Velocity*:

5.1 Peculiar Velocity

Peculiar velocities constitute an unavoidable source of scatter in the observed galaxy redshifts and may, in some cases, affect the classification presented in the *Criterion* column.

In the sample of 700 galaxy pairs analyzed in this work, the mean absolute redshift difference between the two galaxies in each pair is (\Delta z = 7.85 \times 10^{-4}), corresponding to a radial velocity difference of approximately 235 km s(^{-1}).

This value is remarkably close to the typical peculiar velocities of galaxies (approximately 200–300 km s(^{-1})).

Consequently, whenever the expected redshift difference between the two galaxies—arising from differences in stellar mass or mean stellar age—is small, a favorable peculiar velocity may increase or decrease the observed redshift of one galaxy sufficiently to reverse the relative ordering of the pair, thereby changing the value assigned to the *Criterion* column.

Conversely, when the predicted redshift difference is significantly larger than the expected contribution from peculiar velocities, the classification is expected to remain unchanged.

Therefore, peculiar velocities act primarily as a source of statistical noise for galaxy pairs with nearly identical redshifts, reducing the significance of some individual cases without introducing a systematic bias into the overall analysis.

Peculiar velocities can reduce the observed agreement between the predicted and measured redshift ordering by randomly reversing the classification of some galaxy pairs, but they cannot systematically increase the agreement. Consequently, any statistically significant correlation detected despite this additional source of scatter should be regarded as a conservative estimate of the underlying effect predicted by the proposed model.

5.2 Gravitational Potential Effect

As mass and local density increase, the conventional gravitational redshift occurs: photons lose potential energy while escaping the primary galaxy's deep potential well, stretching their wavelength and increasing the redshift. Simultaneously, within the DUT framework, this same intensified gravitational field accelerates local atomic shrinkage, which decreases the redshift.

Therefore, we observe two intrinsic factors acting in diametrically opposite directions on the primary galaxy's spectrum: the gravitational potential (elevating redshift) and the metric contraction rate (reducing redshift). In older systems, the cumulative temporal factor allows mass-induced shrinkage to predominate over the potential well, shifting the statistical mean in favor of the "Criterion = 1" status, while the competition with gravitational redshift acts as a regulatory mechanism that prevents absolute asymmetry.

6. Conclusion

The analysis of 700 dense galaxy pairs revealed a symmetry-breaking pattern in relative redshifts with a significance exceeding 3-sigma.

This phenomenon finds a natural physical justification within the DUT. The data support the thesis that more massive and older galaxies shrink at slightly faster rates, generating lower intrinsic redshifts.

This work establishes robust observational evidence in favor of matter metric contraction models, suggesting that cosmological reference frames warrant profound revision under the scope of the theory proposed by author.

References:

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